

Health & Safety

Standard Work with concrete



Reference

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The standards presented in this document serve as a practical guide to define the means to implement to ensure the activities related to concrete are performed on a safe manner.

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1. Formworks - Assembly and stability

- The assembly and/or use of the formworks requires training by a competent person.
- The assembly and stability of the formworks are carried out in accordance with the Methods statement, or at least as per the manufacturer's instructions
- The assembly must be formally controlled if a method statement requires it (complex formwork assembly)
- The stability of the formworks is achieved via a calculation note by the Methods
- Prefer the use of metal frame counterweight. Otherwise, "homemade" counterweight are designed by The technical Department (weight, lifting steel type, number of lifting points...)
- Use the additional formwork lifting bars if necessary
- Lateral stability: 1:3 rule for great heights



Working platform
equipped with walkway
+ access + collective
protection on all sides



Metal frame
counterweight



Installation of raised
formwork



HUS screw (Evolution 3) reusable if checked by
using the template.
Otherwise reuse is prohibited.

2. Formworks – Fall prevention and access

- The formworks are equipped with walkways and access
- The lateral gates are "closed" each sides
- The guardrails on the formwork sides are reassembled before removing the second formwork

Guardrails on the
formwork side



3. Formworks width < 60cm

These formworks are systematically coupled to a guide of formworks.

Vertical storage of forms < 60cm prohibited
Ground storage required



4. Formworks width < 120cm

Number of bracing pull/push props and counterweight to be put in place: see manufacturer documentation or Method plans

Stabilization of B07/B20 formworks documentations (with compass or with braces) are available among The Plant/Material Department.



5. Decompression boxes

- The decompression boxes allow effortless release of the rod. As soon as there is a superposition of formwork, the lowest level can be equipped with them.
- Follow the manufacturer's provisions depending on the different formwork assembly configurations.



6. Flat slab floor/plywood formworks

Flat slab floor/plywood formworks are sized by the Methods Department. Formworks reception and removal agreement are formalized.

Shoring towers must include:

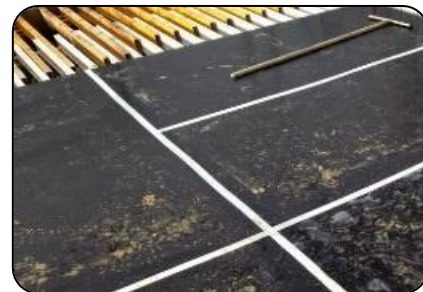
- A complete workfloor with hatch access at the top of each tower
- Guardrails at the top level
- An access ladder attached to the tower
- Planks (**equipped with anti-lift protection**) at intermediate level(s)
- Bracing guaranteeing the stability of the towers from assembly to dismantling

Reduce the deployment of the adjustable jacks as much as possible (limits the access height).

Verification and acceptance are carried out by a competent person (Temporary Works Engineer).

The formwork removal phase is anticipated during the preparation of the operating mode.

Prioritize the use of plywood with dimensions that allows a correct handling by a single operator.



Traditional plywood formwork

Prior any works on the working platforms :

- The guardrails are in place on the edge walls, and at the end of the formwork beams (joist slides)
- Access to the flat slab floor by staircases or escalib are planned and anticipated



Joist slider for fall protection



Shifting shoring towers trolley

Access by Escalib



7. Modular formworks

Types of modular formworks

There are several types of modular formworks :

- Steel formwork (difficult because heavy to implement)
- Aluminum lightweight formwork
- Composite lightweight formwork requiring few tools

Contact your Material Department for any request for the use of these modular formworks



Steel modular formwork



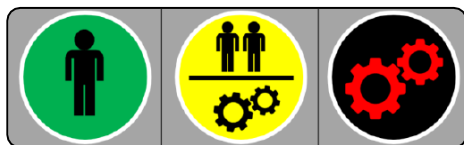
Aluminium modular formwork



Composite modular formwork

Vigilance points

- Implementation of collective protections and necessary access, without any improvisation
- Weight/ manuel handling:
 - <15 kg = handling by one person
 - Between 15 and 30 kg = handling by 2 people or mechanized
 - > 30 kg = Mandatory mechanized handling



Integrated guardrails

- Suitable lifting points, either integrated into the formwork, or available to the user
- Bracing: Compliance with Methods/Manufacturer requirements
- Packaging: ensure that unsealing does not modify the stability of the package and its contents.

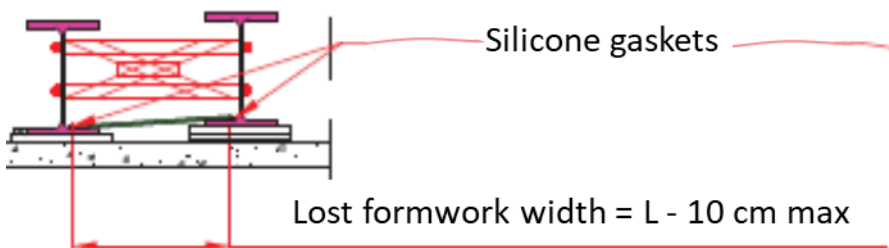


Bracing

8. Lost formworks

VIROC® formworks

VIROC® formworks are permanent lost formworks, generally used between metal or concrete beams during the construction of bridges, viaducs...



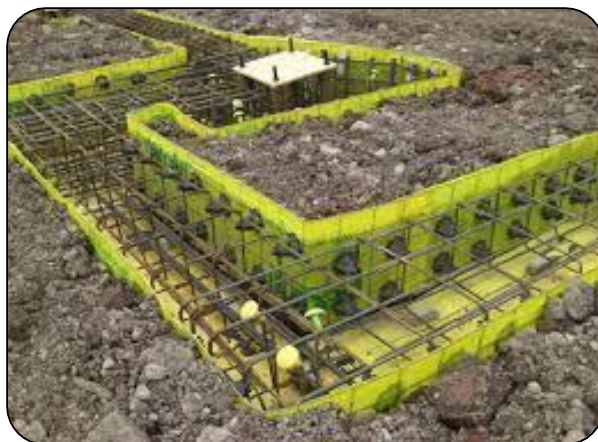
VIROC®

- Order VIROC® formworks to the right size (pre-cut in the factory)
- The “stay-in-place” of the VIROC® plates in between beams is validated by the Methods

Other examples of lost formworks



Cardboard formworks
for columns



Pecafil formworks

9. "Traditional" timber formworks

"Traditional" timber formworks

The design of these formworks provides:

- The intrinsic structure of the formwork and its stability at all phases of assembly/use/disassembly
- Compliant collective protections
- Proper access to the workplace
- Formwork lifting with suitable and compliant lifting devices
- Concreting method
- Formwork removal (formwork removal key, etc.)

No crowbar for formworks removal

- Collective protections installed on wooden support must be designed by the Methods Department



Plate for guardrails used on wooden support

- If formworks are handled by a crane, lifting points are foreseen (clearly identified and dimensioned)



The lifting plan is validated and compliant



10. Specific formworks

Design phase

From the formwork design phase:

- Favor access by stairs when technically possible
- Tilt the ladders as far as possible
- Provide lockable hatches
- Ensure barrier-free circulations for workers
- The work floors are resistant/watertight (junctions, mud flaps, etc.)
- Guarantee easy access to the different platform levels (elevator, stairs between platforms, etc.)



Mandatory ergonomic study in the design phase
(Health & Safety Department)

Safety (and ergonomics) in Design = production gain
Correction during operation phases = additional costs and time loss

Use of formworks

Production phase:

- Respect the implementation procedures and check the anchors at the different phases
- The anchoring devices are implemented via removable platforms
- A handling crossbar is used to ensure the safe installation of the anchor shoes (sizing and validation methods)
- Do not release the crane until the anchor shoe is locked
- Validate the control sheet before any hoisting operation
- Take into account weather conditions (wind) for lifting



Sealing flaps



Climbing formworks

11. Precasting on site / foreign precast

Precasting on site

- Reinforcement plan validated by the Technical Department
- Selfverification before concreting
- Compliance with the minimum strength of concrete prior formwork removal
- Formwork plates are equipped with collective protection when necessary



Warning: In case of climatic conditions, particularly in cold weather, risk of lack of curing of the concrete (particularly specific concrete like low carbon ones...)



Precast beams bench

Foreign precast

- Mandatory supplier visit by the team to ensure proper compliance with the specifications (reinforcement, lifting points...)
- Marking of lifting points
- Layout of sheaths for collective protections (compliance with spacing rules between posts)



Precast storage

- Installation of wedges (wooden beams for example) between each precast flat stored (number and position of wedges to be validated according to the manufacturer's instructions or the Technical Department)
- Maximum stacking number must be defined
- Max height of slinging points = 1.50m



Segments storage



Supplier's storage rack
Permanent slings on precast



12. Concrete pumps – 1/2

Ground stability

- Pump trucks: check the stability of the ground BEFORE installation on site (presence of embankments, loose earth, underground networks, etc.) - Park brake must be activated
- Carry out soil bearing capacity if necessary and know the descent of load on each pump truck skid
- The aerial and buried networks are known and tackled with

No one near the pump during priming, starting and cleaning
PPE must be worn at all times!



Pump mixer truck
(or mixo-pump or Pumi)



Pump truck + “disassociated”
concrete mixer



Concrete pump

Starting the pump

- Priming the pump with cement slip
- Take into account site effects (storm, wind), possible coactivity with other lifting equipment - Mark the work area
- Check hoses, couplings and clamps
- Anticipate manual handling of constituent elements (pump pipes, etc.), by proposing helping system for manual handling (trolley, rollers...)
- Grounding required when working near overhead/buried power lines in addition of a power detection device



Mandatory
distribution plates

- The pump attendant is always in visual or radio contact with the concreting team
- The pump attendant always signals the priming, the start of pumping, and each restart (audible signal)
- Ensure that the pump hopper is always full (minimum 80%, excluding the end of implementation).



ANOMALY = IMMEDIATE STOP

13. Concrete pumps – 2/2

Concrete hoses supply

- All hose fitting is equipped with seal + collar + pin + strop
- Never put a metal connector at the end of the hose
- Secure the pipes to the ground on stable and fixed supports
- Install an S-connector with handle at the end of the hose:
 - Reduces concrete falling
 - Ensures better ergonomics at the workstation



Cleaning of pump and pipes on the ground

In any cases, keep the pump stabilized at all time, including when cleaning the pumping group and the tank, with the boom folded.

Cleaning of the concrete pump boom piping:

- Mandatory suction washing of the ball from the end of the boom towards the bin

Cleaning of the floor piping:

- Use a water washing unit, or bentonite

Cleaning of the hopper:

- Hopper: necessarily equipped with an agitator cut in case of opening of the grid

Do not throw away leftover concrete.

Cleaning concrete pipes with compressed air is prohibited!

Spreaders and concrete masts

Benefits :

- Good stability of the machine even under intensive pumping
- Easy concrete pouring
- Easy to handle
- Elimination of handling and installation of concrete pipes on the ground for works under slabs and inaccessible with traditional spreading arm



Concrete spreader

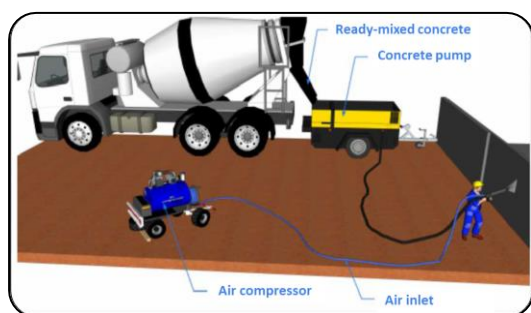
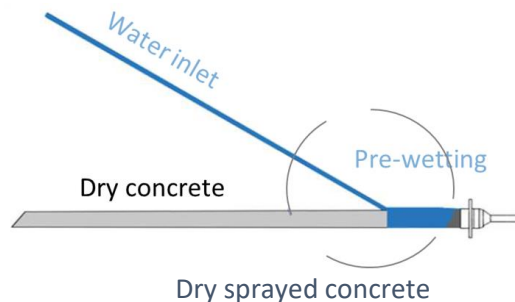
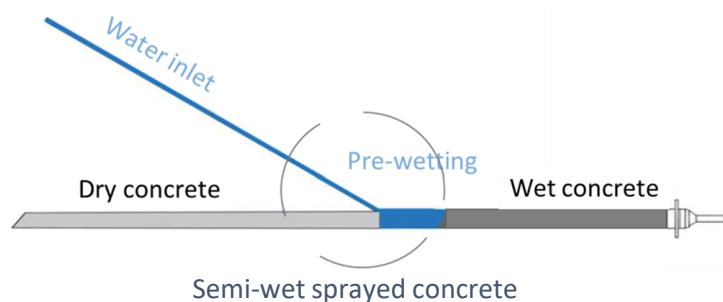


Concreting mast

14. Shotcrete / Sprayed concrete

3 techniques generally used to spray concrete:

- Dry process
- Semi-wet process
- Wet process



Preventive maintenance is planned for the entire installation during design:

- Type of pipe and thickness of steel (very corrosive cement)
- Positioning of fittings for possible replacement (accessibility)
- Silo: clogging control, hygrometry, valves, etc.



Access to areas during concreting is prohibited for unauthorized personnel.

The workers are trained in the use of the equipment and wear specific PPE :

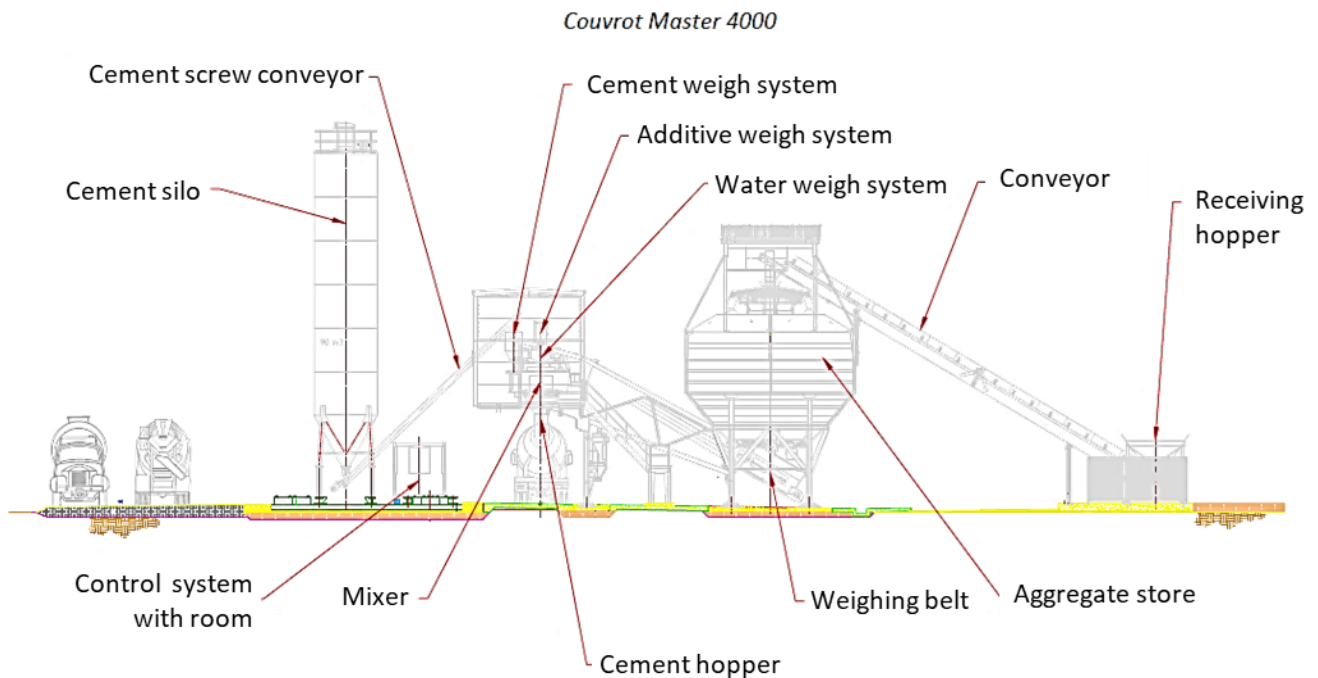
- Ventilated mask equipped with the appropriate filter for frontline personnel (or equivalent protection)
- Respiratory protection if necessary for support personnel



15. Concrete plants

The rental and assembly of concrete plants are carried out by the Vendor company / Material:Plant Department which makes it available, or even operates it as a subcontractor.

- Check the bearing capacity of the land
- Comply with the Site Installation Plan and vehicle and pedestrian circulations :
 - Arrival and storage of aggregates
 - Arrival/departure of mixer trucks
- Limit reverse circulation as much as possible
- Ensure the declaration of classified installations for environmental protection, when applicable in local regulations
- Mandatory presence of emergency stop systems (regular tests)
- Compliance with the Lock out/Tag out procedure(s) for any assembly, maintenance or repair, and dismantling interventions
- User personnel are trained in the operation and use of the plant



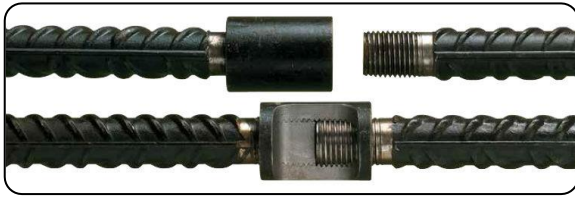
Standard diagram of a concrete plant

Pedestrian circulations and interventions are prohibited within the perimeter of use of the plant without agreement from the plant manager

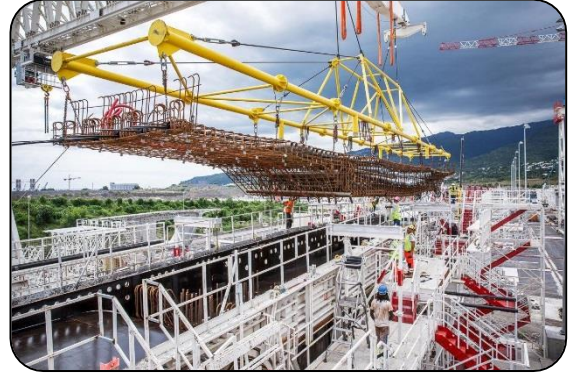
16. Reinforcement – 1/2

Foreign reinforcement frames (Cut Shaped & Assembled)

- Lifting points are defined, dimensioned and identified
- Prioritise the use of a spreader when lifting
- If necessary, foresee inherent slings
- Unloading of such reinforcements is anticipated (slinged from the bottom)



Rebar coupler



Spreader

Reinforcement frames (Cut Shaped)

Cut-Shaped reinforcements are obtained by cutting and shaping rebars on demand. They are assembled on site.

- Lifting points are identified
- If necessary, provide permanent slings
- If the rebars are moved, apply the rules for lifting
- Provide collective protection for assemblies at height, as well as access to the workstation



17. Reinforcement – 2/2

Securing circulation on reinforcements

Circulation on the reinforcement cages is defined and protected :

- 10x10cm mesh size by design
- Additional welded mesh 10x10cm mesh size
- Circulation mat on the upper part of rebars
- Any dispositions that allow main safe access



Circulation mat

Work inside reinforcing cages

Before being authorized to access the interior of the rebar cages, a specific risk analysis must define:

- The possibility of installing a decking
- The possibility of implementing collective protections
- Use of double lanyard safety harness if necessary with adequate and compliant anchor points

Specific procedure :

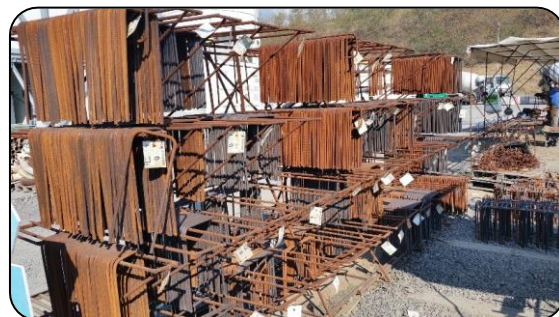
- Consider postural constraints at the design stage
- Plan emergency response



Worker inside rebar cages

Manual handling of rebars

- The packs of rebars are stored as close as possible to the precasting area
- If necessary, use trestles, in order to be able to work at man's height



Reinforcement preparation workshop



Ligation gun

18. Repair works

Passivation of steels

Operations prior to passivation:

- Exposing steels
 - Referenced lightened jackhammer / vaccum at source if possible
 - Use of vibrating tools = 2hrs/day/worker
- Wear specific PPE depending on the tasks to be carried out



Passivation of steels:

- Wear specific “Chemical risk” PPE

Leveling with post-passivation mortar:

- Wear specific “concrete” PPE



Rebars to passivate

Surfacing / Sanding

In order to limit the penibility:

- Use ergonomic sanding tools with vacuum at source
- The heights of the working floors are appropriate for the tasks to be carried out :
 - Scaffolds : check access, circulation, weight capacity and supply of equipment (ensure that the worker can walk upright)
 - Roofing brackets: validation by the Methods, access to be provided
- Wear specific PPE depending on the tasks to be carried out
- Ensure the admissible load of the scaffolding platforms (weight of equipment + operators) - Same for basket



Sanding cart

Installation of carbon strips

Carbon fibers are used to reinforce the resistance of degraded structures

- Wear specific “chemical risk” PPE
- Respect the conditions for implementation of carbon sheets (hygrometry, outside temperature, etc.)
- Provide tools to assist the transport of materials
- Means of work at height access are anticipated



Installation of carbon fabrics

19. Jacking / Unjacking

Preparation of operations

The heights of the working floors are appropriate for the tasks to be carried out.



Use of scaffolding:

- Check access, circulations and supply of equipment (ensure that the worker can walk around standing)
- Ensure the admissible load of the scaffolding platforms (weight of equipment + operators) - Same for basket

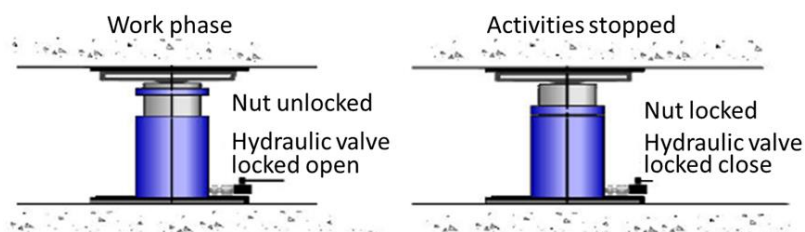
Brackets: validation by the Technical Department, access to be provided

NB : Provide tools to help with materials handling (trolleys, jib, etc.)

Jacking phase

In order to limit the penibility:

- Ensure the flatness of the jacking surfaces
- Ensure compliant hydraulic connections
- Provide tools to help with materials handling (trolleys, crane, etc.)
- Ensure the capacity and conformity of the adjustable jacks
- Permanent contact between the operator and the teams



Adjustable jacks equipped with safety nuts

20. Pre/post-tensioning

Trainings

- Pretensionning Works Director
- Pretensionning Implementation Manager
- Pretensionning Operator

Standard operating procedure

Threading cables into conduits (threading machine) :

- Mechanize the handling of reels in the reels
- Mark the reel area and remain vigilant when opening the reels (strips)
- Sheath the strand to the machine as soon as it exits the reel, and from the machine to the threading nose
- Do not stand in front of the strand exit / monitor the reel constantly
- Cut strands with hydraulic shear/electric grinder

Installation of anchors and jacks:

- Compliant hoists and slings

Tensioning of cables:

- Compliant hydraulic cylinder and pumps
- Prohibition to stay behind the cylinder
- High Pressure verified and guaranteed hoses
- Tensionning Manager present
- Suitable counterweight trolley

Injection of cables with wax:

- Wearing specific anti-burn PPE

Injection of cables with grout:

- Mixer equipped with safety devices (grid with safety switch, emergency stop, etc.)
- Hoses in perfect conditions
- Wear specific "concrete" PPE (chemical hazard)
- Mechanize the handling of cement bags

Examples of tensioning



External tensioning



Internal tensioning



Additional tensioning



Positioning the cylinder



Example of threading by pushing



Injection point with pressure control